THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 12: **Surface Areas and Volumes** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The radii of two solid spheres are in the ratio 4 : 7. The ratio of their surface areas is: 1

(a) 16 : 49 **(b)** 64 : 343 **(c)** 32 : 49 **(d)** 4 : 7

2. The radii of two solid spheres are in the ratio 1 : 5. The ratio of their volumes is: 1

(a) 1 : 125 **(b)** 1 : 25 **(c)** 1 : 5 **(d)** 2 : 125

3. A solid cylinder of base radius 6 cm and height 8 cm is melted and recast into identical solid cones, each of base radius 2 cm and height 12 cm. The number of cones obtained is: 1

(a) 22 **(b)** 18 **(c)** 36 **(d)** 6

4. A hemispherical bowl has outer radius 10 cm and inner radius 2.5 cm. The volume of material used to make the bowl is: 1

(a) 4125 cm^3 **(b)** 2062.5 cm^3 **(c)** 6187.5 cm^3 **(d)** 1031.25 cm^3

5. The largest possible solid sphere is carved out of a cube of edge 84 cm. The volume of the material wasted is: 1

(a) 437472 cm^3 **(b)** 310464 cm^3 **(c)** 592704 cm^3 **(d)** 282240 cm^3

6. A solid sphere of radius 42 cm fits exactly inside a cylinder (it touches the curved surface and both bases). The volume of the space in the cylinder not occupied by the sphere is: 1

(a) 465696 cm^3 **(b)** 155232 cm^3 **(c)** 310464 cm^3 **(d)** 776160 cm^3

7. A solid cylinder of radius 10 cm and height 3 cm is melted and recast into a cone of the same base radius. The height of the cone is: 1

(a) 1 cm **(b)** 6 cm **(c)** 3 cm **(d)** 9 cm

8. A solid cone and a solid hemisphere have equal base radii 21 cm and equal volumes. The height of the cone is: 1

(a) 42 cm **(b)** 21 cm **(c)** 31.5 cm **(d)** 63 cm

9. **Assertion (A):** A solid is made by placing a cone on a hemisphere of the same radius. The total surface area of the solid is the sum of the curved surface areas of the cone and the hemisphere. 1
Reason (R): The base of the cone and the flat face of the hemisphere are joined together, so neither is exposed.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The volume of a sphere of radius 3 cm is 36π cm^3. 1
Reason (R): The surface area of a sphere of radius r is $2\pi r^2$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A solid metallic sphere of radius 12 cm is melted and drawn into a wire of circular cross-section of radius 1 cm. Find the length of the wire. 2

12. A solid sphere of radius 15 cm is completely dropped into water in a cylindrical vessel of base radius 10 cm, and no water spills. Find the rise in the water level. 2

13. A hemispherical bowl has outer radius 9.5 cm and inner radius 5 cm. Find the volume of material used to make the bowl. 2

14. A solid metallic sphere of radius 6 cm is melted and drawn into a wire of circular cross-section of radius 1 cm. Find the length of the wire. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A solid metallic cylinder of base radius 9 cm and height 48 cm is melted and recast into solid spheres of radius 3 cm. Find the number of spheres obtained. 3

16. A solid metallic cone of base radius 20 cm and height 10 cm is melted and recast into a solid sphere. Find the radius of the sphere. 3

17. A solid cylinder of base radius 14 cm and height 6 cm is melted and recast into identical solid cones, each of base radius 7 cm and height 18 cm. Find the number of cones obtained. 3

OR

A solid metallic cylinder of base radius 6 cm and height 54 cm is melted and recast into solid spheres of radius 3 cm. Find the number of spheres obtained. 3

18. A solid metallic cone of base radius 2 cm and height 8 cm is melted and recast into a solid sphere. Find the radius of the sphere. 3

OR

A solid cylinder of base radius 9 cm and height 20 cm is melted and recast into identical solid cones, each of base radius 3 cm and height 15 cm. Find the number of cones obtained. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A solid metallic sphere of radius 6 cm is melted and recast. (Leave answers in terms of π .) 5
(i) Find the volume of the sphere.
(ii) Find the height of a solid cylindrical rod of radius 4 cm that can be made from it.
(iii) How many solid spheres of radius 3 cm can be made from the original sphere?
(iv) Compare the total surface area of these small spheres with that of the original sphere.

20. A tent is a cylinder of radius 21 m and height 15 m surmounted by a cone of the same radius and height 28 m. Find the area of canvas (in m^2) needed to make the tent. 5

OR

The largest possible solid sphere is carved out of a cube of edge 147 cm. Find the volume of the material wasted. 5

END OF QUESTION PAPER **Total: 40 Marks**